

20. ROS2 Recording and Playback Tool

Note: The accompanying reference video uses the ROS Humble version, for reference only

1. Recording and Playback Tool Overview

A commonly used recording and playback tool in ROS2 - bag2. This tool is used to record topic data. We can use this command to store topic data as a file. Later, we don't need to start nodes, and can directly publish topic data from the bag file.

This tool is very useful when working with a real robot. For example, we can record a segment of topic data when a problem occurs with the robot. After recording, we can publish it multiple times for testing and experiments. We can also share topic data with others for algorithm verification, etc.

Let's try using the bag tool to record topic data and replay it.

2. Usage Tutorial

2.1. Start the Topic Node to Record

Like talker in ros2 demo:

```
ros2 run demo_nodes_py talker
```

2.2. Record

`/topic-name` is the topic name:

```
# Record a single topic
ros2 bag record /topic-name
# Record multiple topics
ros2 bag record topic-name1 topic-name2
# Record all topics
ros2 bag record -a
```

Other options:

- `-o name`: Custom output file name

```
ros2 bag record -o file-name topic-name
```

- `-s`: Storage format

Currently only supports sqlite3, others are still being expanded.

2.3. View Recorded Topic Information

Before playing a video, we can view video-related information such as topic recording time, size, type, and quantity:

```
# Assuming the recorded file is rosbag2_2023_10_31-07_58_23
ros2 bag info rosbag2_2023_10_31-07_58_23
```

2.4. Play and View

2.4.1. Play

Next, we can replay the data. Use the following command to play data:

```
ros2 bag play rosbag2_2023_10_31-07_58_23
```

2.4.2. View

Use ros2 topic commands to view data:

```
ros2 topic echo /chatter
```

2.4.3. Playback Options

1. Speed playback -r

The -r option can modify playback speed. For example, -r 10 means 10x speed playback:

```
ros2 bag play rosbag2_2023_10_31-07_58_23 -r 10
```

2. Loop playback -l

This is for single-track loop:

```
ros2 bag play rosbag2_2023_10_31-07_58_23 -l
```

3. Play a single topic:

```
ros2 bag play rosbag2_2023_10_31-07_58_23 --topics /chatter
```

3. Examples

3.1. Run talker Node

```
ros2 run demo_nodes_py talker
```

3.2. Record

```
# Record all topics  
ros2 bag record -a
```

```
root@unbutu:~# ros2 run demo_nodes_py talker  
[INFO] [1698740480.382000661] [talker]: Publishing: "Hello World: 0"  
[INFO] [1698740481.364410896] [talker]: Publishing: "Hello World: 1"  
[INFO] [1698740482.364672850] [talker]: Publishing: "Hello World: 2"  
[INFO] [1698740483.364670570] [talker]: Publishing: "Hello World: 3"  
[INFO] [1698740484.364849226] [talker]: Publishing: "Hello World: 4"  
[INFO] [1698740485.365054922] [talker]: Publishing: "Hello World: 5"
```

```
root@unbutu:~# ros2 bag record -a  
[INFO] [1698740481.324354588] [rosbag2_storage]: Opened database 'rosbag  
2_2023_10_31-08_21_21/rosvbag2_2023_10_31-08_21_21_0.db3' for READ_WRITE.  
[INFO] [1698740481.339396600] [rosbag2_transport]: Listening for topics.  
..  
[INFO] [1698740481.341651790] [rosbag2_transport]: Subscribed to topic '  
/rosout'  
[INFO] [1698740481.343500660] [rosbag2_transport]: Subscribed to topic '  
/parameter_events'  
[INFO] [1698740481.345063472] [rosbag2_transport]: Subscribed to topic '  
/chatter'
```

How to stop recording? We can directly interrupt recording using `Ctrl+C` in the terminal.

Then you will see a new folder in the terminal named `rosvbag2_2023_10_31-08_21_21`.

Open the folder and you can see the contents:

```
root@unbutu:~# ls  
rosvbag2_2023_10_31-08_21_21  yahboomcar_ros2_ws  
root@unbutu:~# cd rosvbag2_2023_10_31-08_21_21/  
root@unbutu:~/rosvbag2_2023_10_31-08_21_21# ls  
metadata.yaml  rosvbag2_2023_10_31-08_21_21_0.db3  
root@unbutu:~/rosvbag2_2023_10_31-08_21_21#
```

This completes the recording.

3.3. Play and View

Here we play in loop:

```
ros2 bag play rosvbag2_2023_10_31-07_58_23 -l
```

Open another terminal to view the topic:

```
ros2 topic echo /chatter
```

```
root@unbutu:~# ls
rosbag2_2023_10_31-08_21_21_yahboomcar_ros2_ws
root@unbutu:~# ros2 bag play rosbag2_2023_10_31-08_21_21/ -l
[INFO] [1698740802.654185287] [rosbag2_storage]: Opened database 'rosbag2_2023_10_31-08_21_21//rosbag2_2023_10_31-08_21_21_0.db3' for READ_ONLY.
```

```
root@unbutu:~# ros2 topic list
/chatter
/parameter_events
/rosout
root@unbutu:~# ros2 topic echo /chatter
data: 'Hello World: 18'
----
data: 'Hello World: 19'
----
data: 'Hello World: 20'
----
data: 'Hello World: 21'
----
data: 'Hello World: 22'
----
data: 'Hello World: 23'
----
data: 'Hello World: 24'
----
data: 'Hello World: 25'
----
data: 'Hello World: 26'
----
data: 'Hello World: 27'
----
```